# Module 4: Leverage

- Leverage: betting on a spread
- Consequence of debt

  Lecture 12: Value at risk

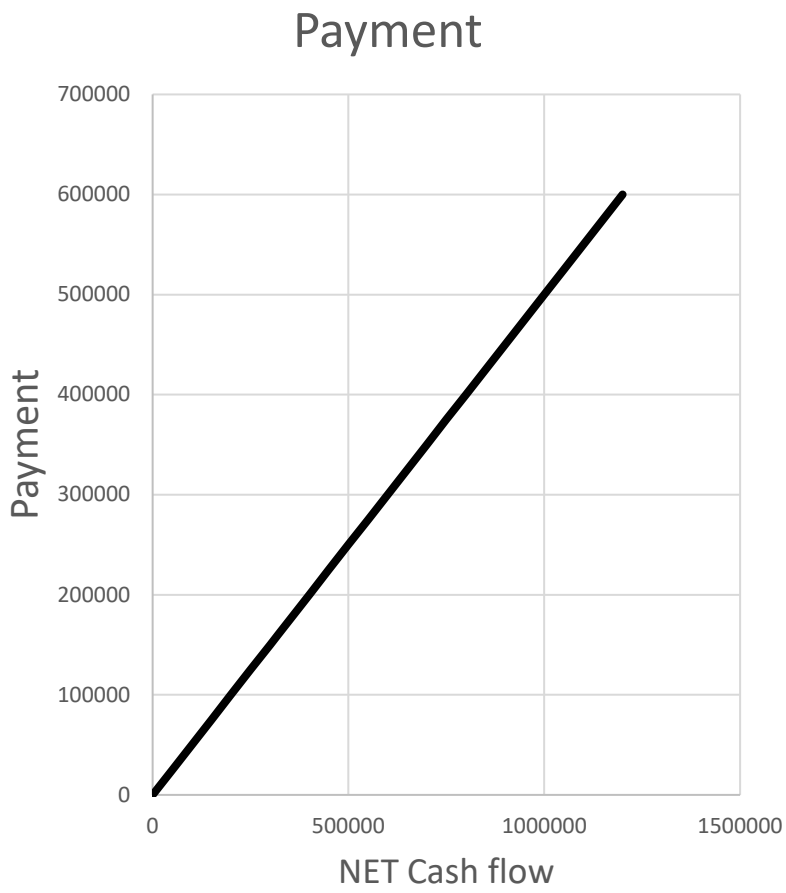
  Lecture 13: Real Estate finance

  Lecture 14: Bank regulation

  Lecture 15: Federal Reserve

Goal is understanding managing non-linear risk.

# Equity vs Risky Debt



BEM 103

# Class 12: Value at Risk and conditional value at risk

**Beyond expected value**

**Financial planning when the world is uncertain**

**Lessons for Long Investors**

**Lessons for Long Funds**

**(leveraged/hedged funds in Lecture 17)**

# Key concept

- Binomial model
- Drift
- Goal of class is to go from pure long to leverage portfolio.
- Value at Risk
  - 5 percentile of the return distribution
  - Could use any other threshold
- Conditional Value at Risk
  - Expected value if return below the VaR

# Beyond Mean-variance

- Mean variance is very convenient
- Setting up a model of asset pricing and thinking about individual preferences and risk taking.
- For short-term completely long portfolios (stock picks) that makes reasonable sense
- In others less so
  - First in financial planning
  - For long intermediaries when investors move funds around when returns are low relative to a benchmark
  - Additional pb when there is leverage

# Financial planning

- Individual wants to retire
  - Has an income desire at retirement (and possibly wants to make bequests) and that translates to a wealth goal.
  - If wealth goal reached, retire
  - If not stay at work (less good)
- So now we have mean-variance with a target
  - Individual wants to know how likely target is met and retire “on time”
  - Important, the target here is like a zero coupon bond that the investor owns.

# The binomial model

- Simple idea:
  - Returns are either good (up x%) or bad (down y%)
  - Probabilities are 0.5 in either case
- Can vary all the assumptions but core of the model has 1 advantage: easy to compute
  - And its limit as the period gets ever smaller is the continuous time model where risk is normal

# Drift

- Start with $V_t$
- If return is good $V_{t+1g} = V_t(1 + x)$
- if return is bad $V_{t+1b} = V_t(1 - y)$
- E$(V_{t+1}) = 0.5V_t(1 + x) + 0.5V_t(1 - y)$
- E$(V_{t+1})$= $V_t\left(1 + \frac{x}{2} - \frac{y}{2}\right)$=>
- E$(V_{t+1} - V_t)$= $V_t\,\frac{x-y}{2}$ => g= $\frac{V_{t+1}-V_t}{V_t}$ = $\frac{x-y}{2}$
- Drift or expected growth rate of V $\mathrm{g} = \frac{x-y}{2}$

# States of the world

- $x > y$ implies g>0
- However, the number of states of the world values of the asset grows at time goes by
- $V_t$ takes on all values defined by $V_1(1+x)^i(1+y)^{t-i} \quad st\ i \leq t$
- And it is not sensitive to the orders of the draw.

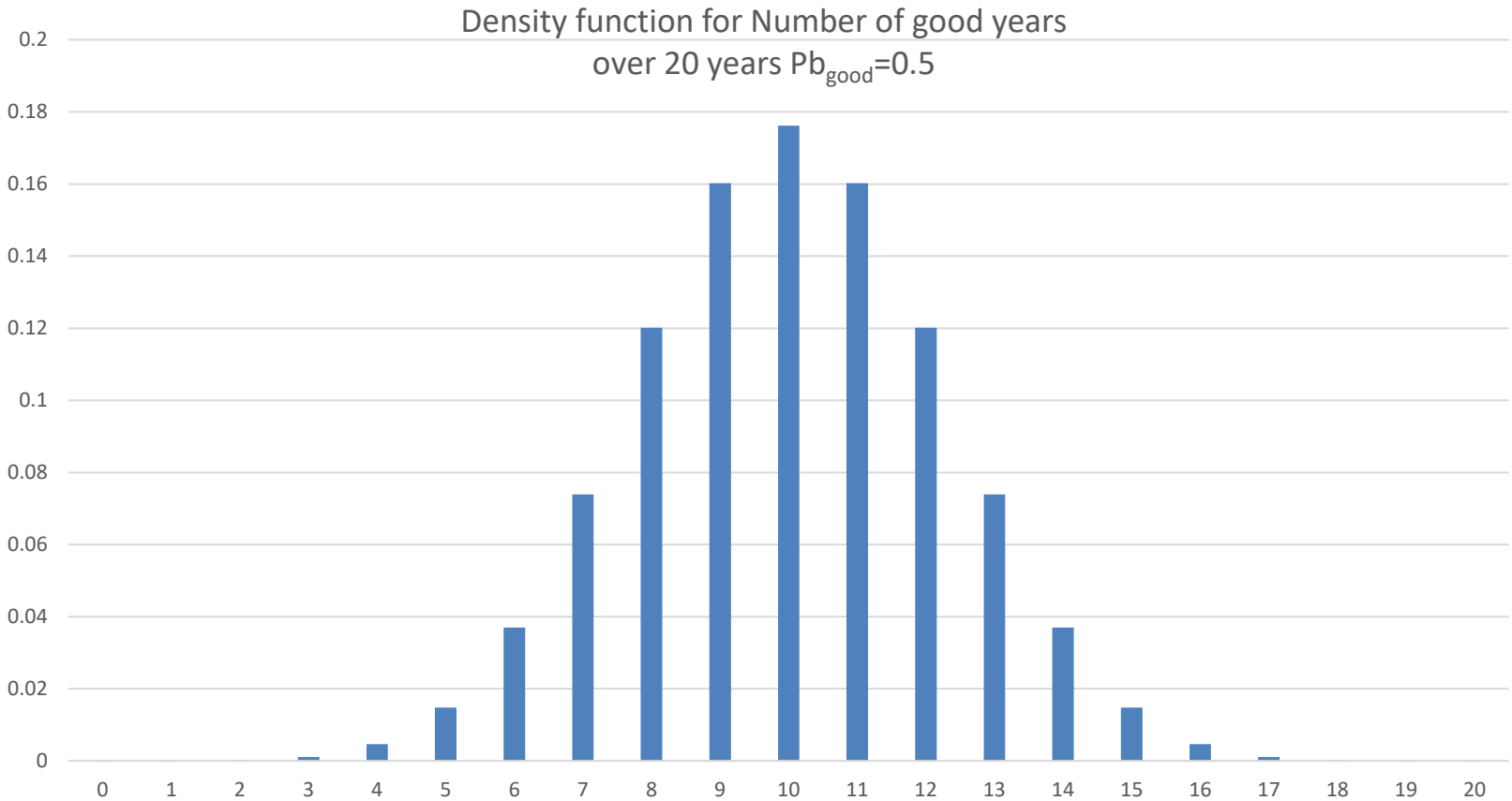
Computing the probabilities is a simple combinatoric pb.

# Example

- Individual has won the lottery and has $1,000,000. Plans to retire in 20 years and works but contemplates not contributing to his retirement program.
- Hires an investment planer who offers him a binary investment plan (good years return 19% bad years lose 2%, expected return 8.5%)
- The expected return over 20 years is 411%= $(1.085)^{20}$-1
  - So the individual expects to have 5.12 million
  - And at 5% an annuity of $255,600
- Is this good enough?

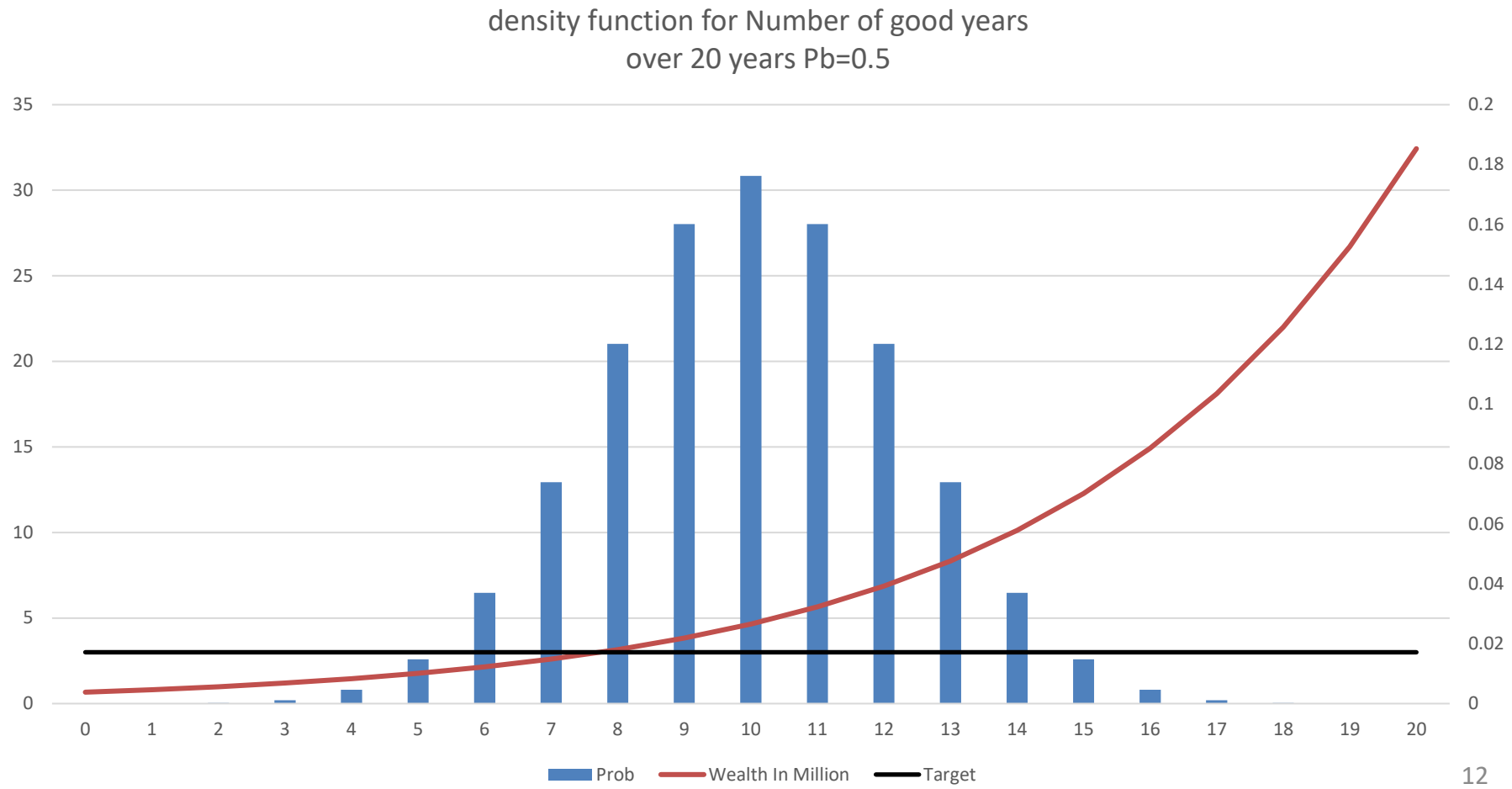
# Of course not!

- Individual wants to retire with at least $150,000 in annual income and this has a wealth target of 3 million.
- Compute the distribution of outcomes (probability of bad years)



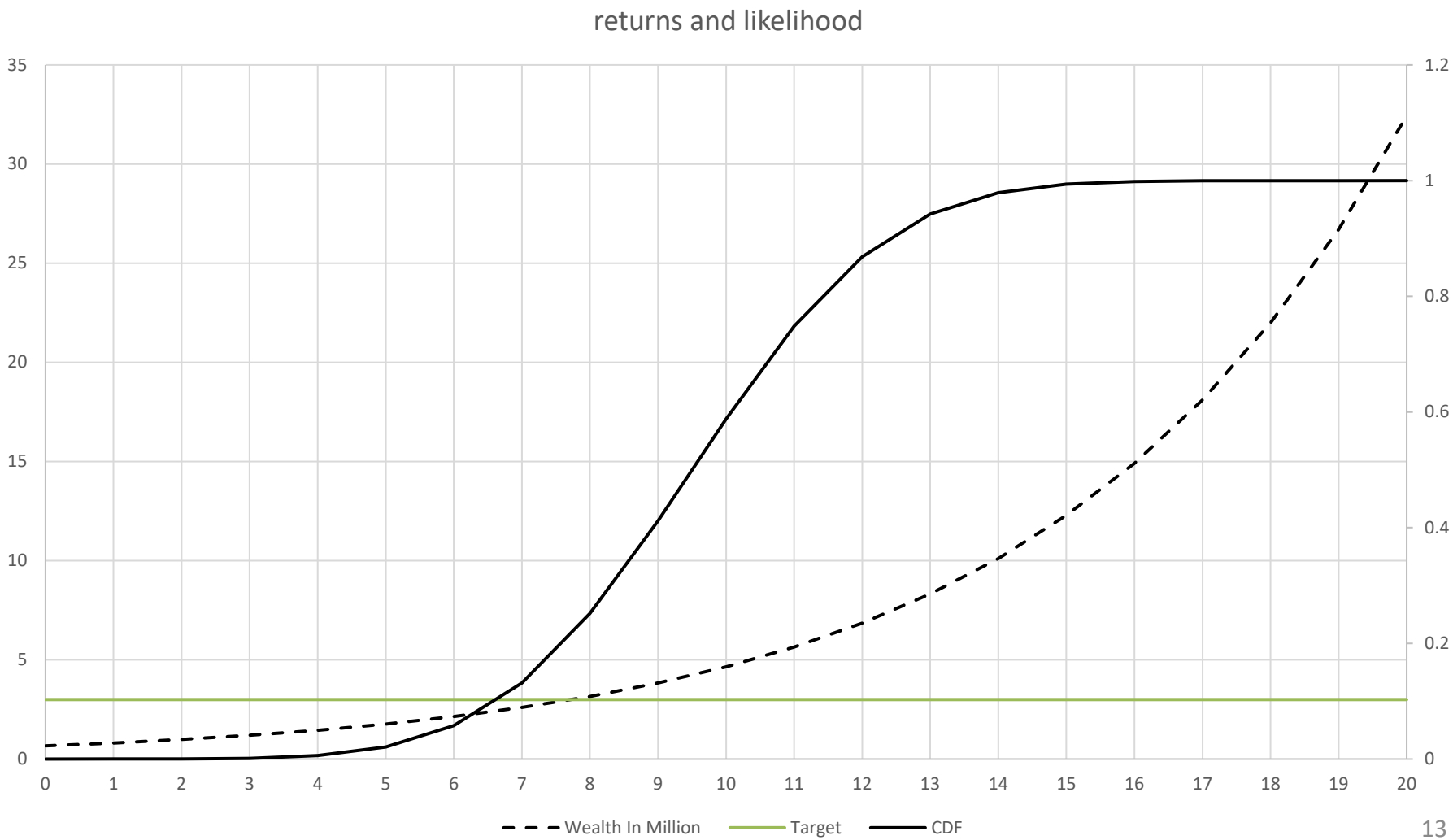
# Of course not!

- Individual wealth target of 3 million.
- Compute the distribution of outcomes (probability of bad years)



Turn the density into a CDF and compute the value of the investment for each realization.

Note with a single investment it does not matter when the bad years happen. All returns are $(1+x)^i(1+y)^{t-i}$



# VaR and CVaR

<u>Value at risk:</u> specific fractile of the distribution

Definition VaR= the fifth percentile 5% of the distribution of outcomes (p.05)

<u>Conditional value at risk:</u>

Definition: Expected value of the program if you are below the VaR.

CVaR=$\frac{\sum_0^N (p_i) V_i}{\sum_0^N (p_i)} sbjt V_i \leq VaR$

Note: the finance convention is the fifth percentile but that is convention only.

The client knows that with 95% probability they will do better than that.

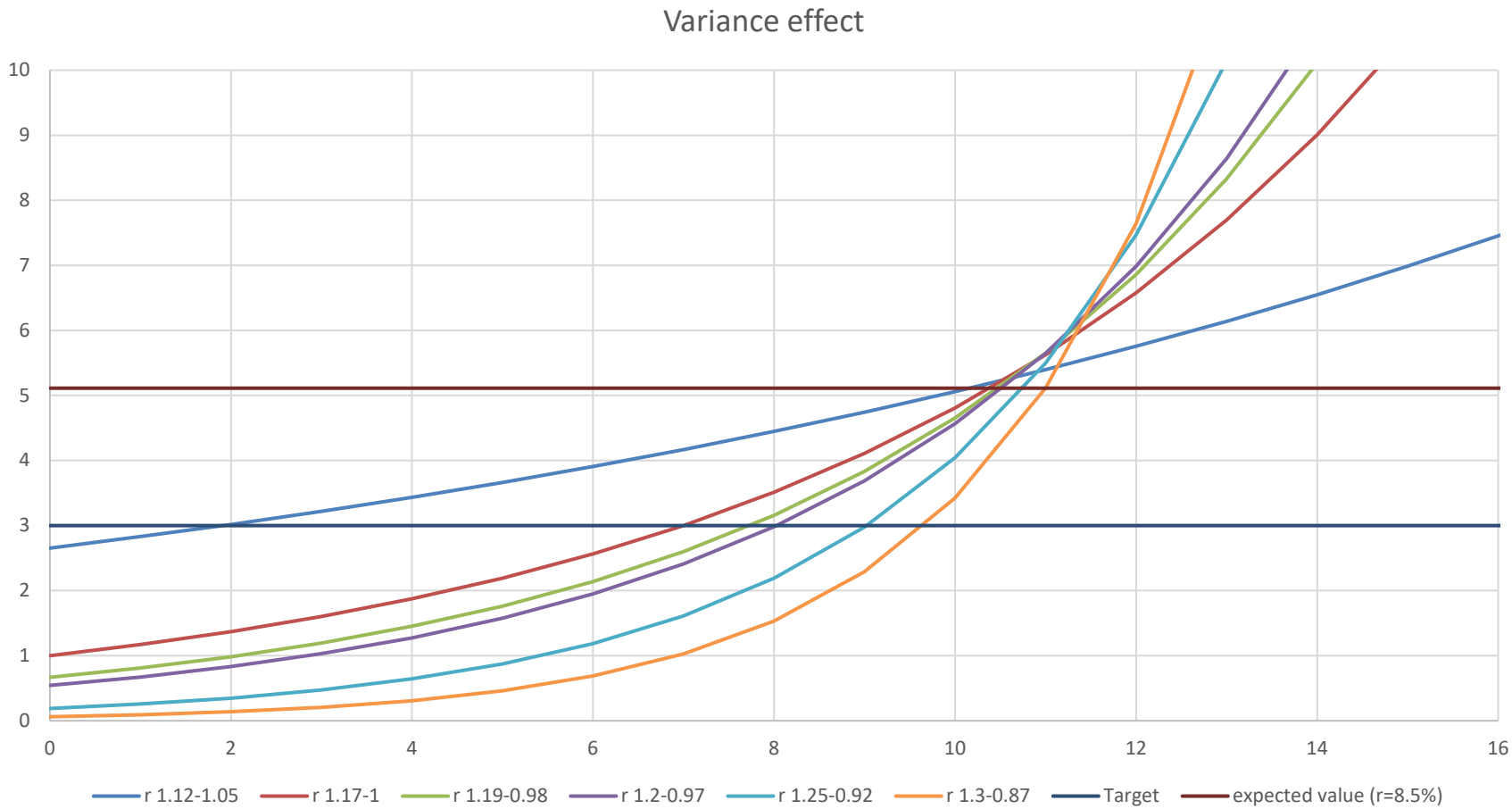
# Target

- Meet 75% of the time yes, 25% of the time no
- Value at risk (25%)=$3 million
- Value at risk (5%)=$2.6 million
- If you want to drive the VaR(5%) up to $3 million (drive down the prob you do not meet the target) you can save
- Computation a bit more tedious because now you are not history independent.

# Value at risk with

- No Savings a year
  - Value at risk (5%)=$2.14M
  - Cvar=$1.68 M
- Savings 10k a year
  - Value at risk (5%)=$2.47M
  - Cvar=$2.24 M
- Savings 30k a year
  - Value at risk (5%)=$2.99M
  - Cvar=$2.82 M

# Risk and return



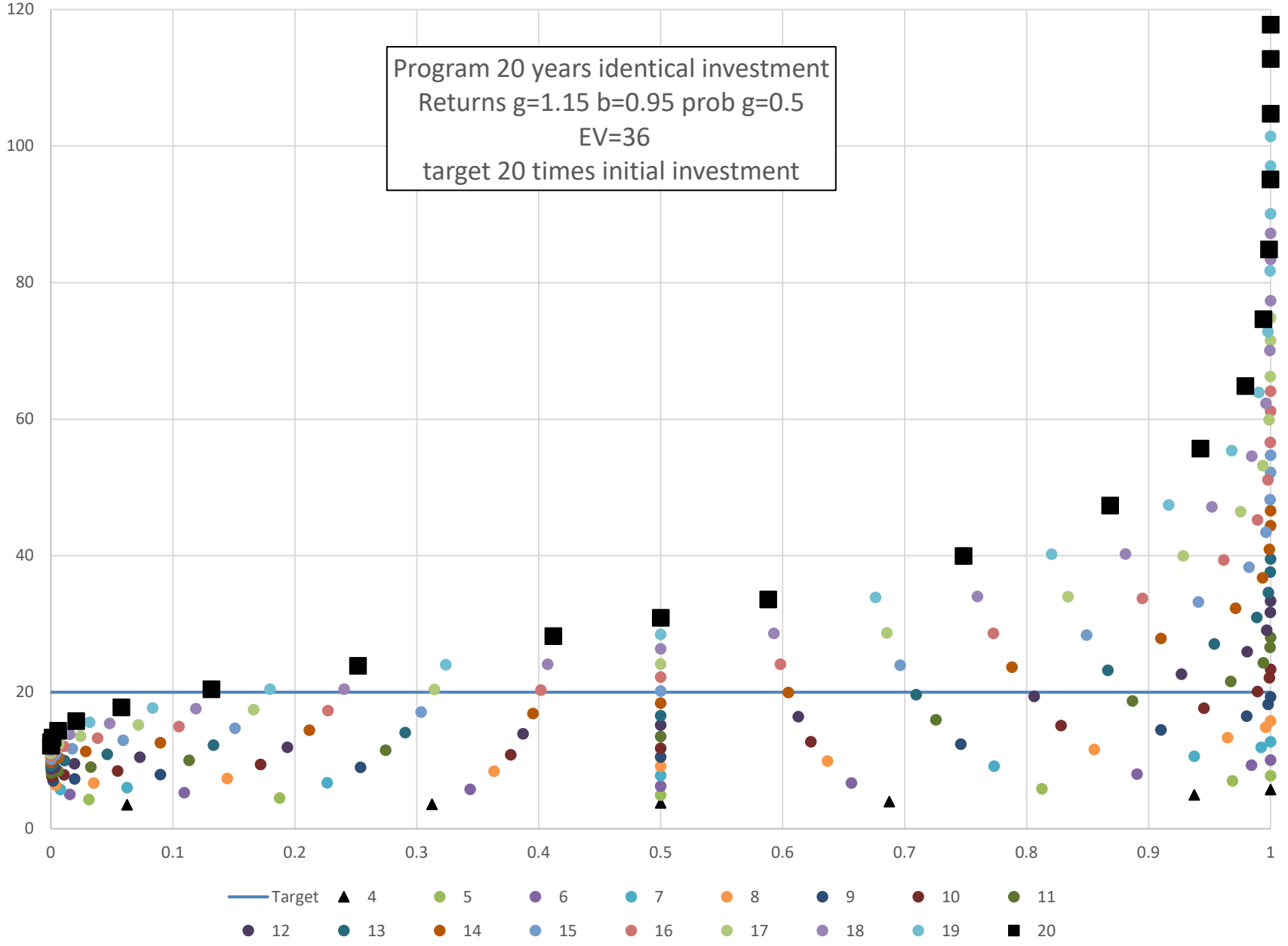

The more variance the more good draws you need to reach the target.
What do you prefer

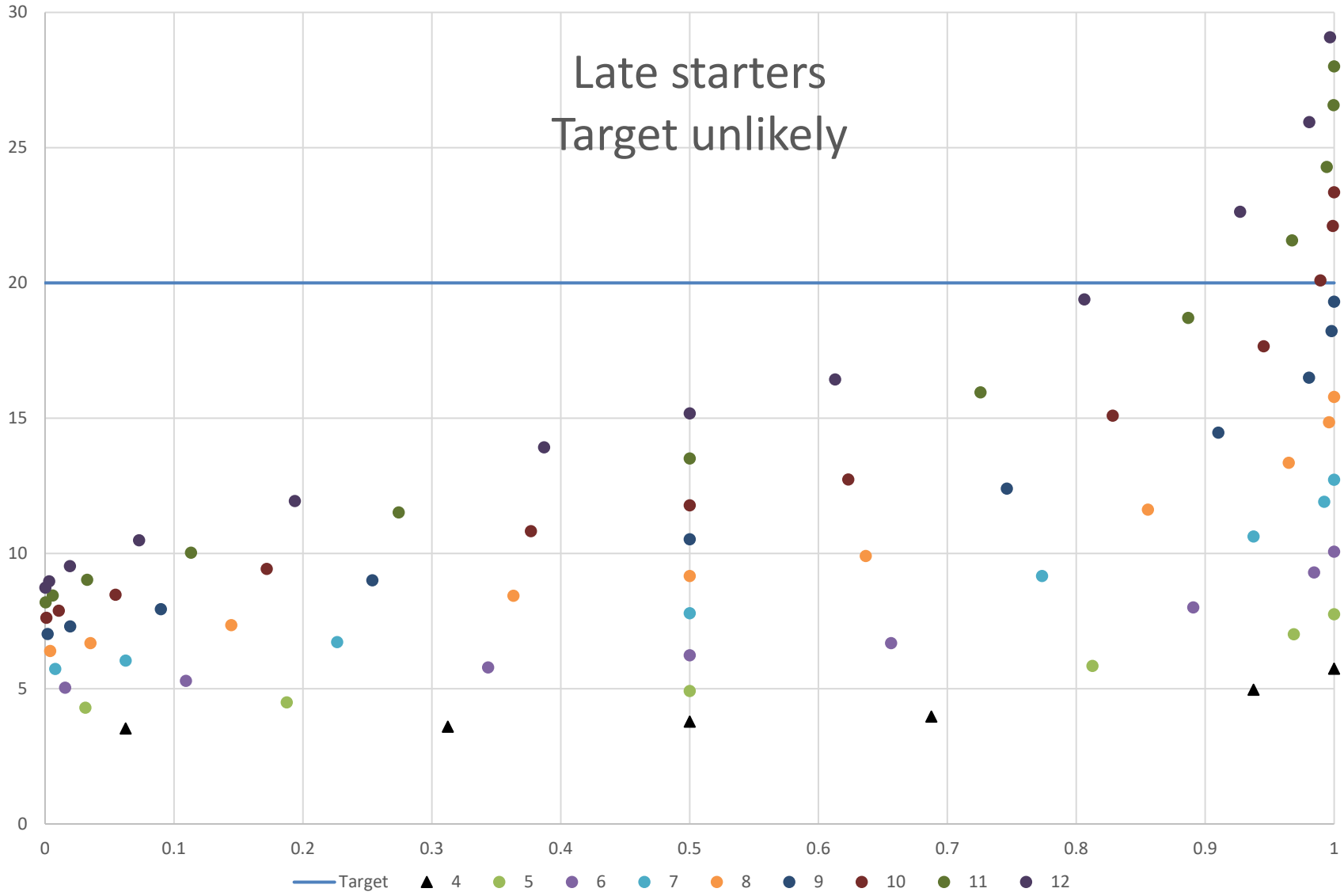
# Example 2

- Individual plans to retire in 20 years and works but contemplates contributing 1 unit to retirement program each year for 20 years.
- Hires the same IP who offers him the same investment plan that is binary (good years return 15% bad years lose 5%, expected return 10%)
- The individual has a target of 20 units wealth

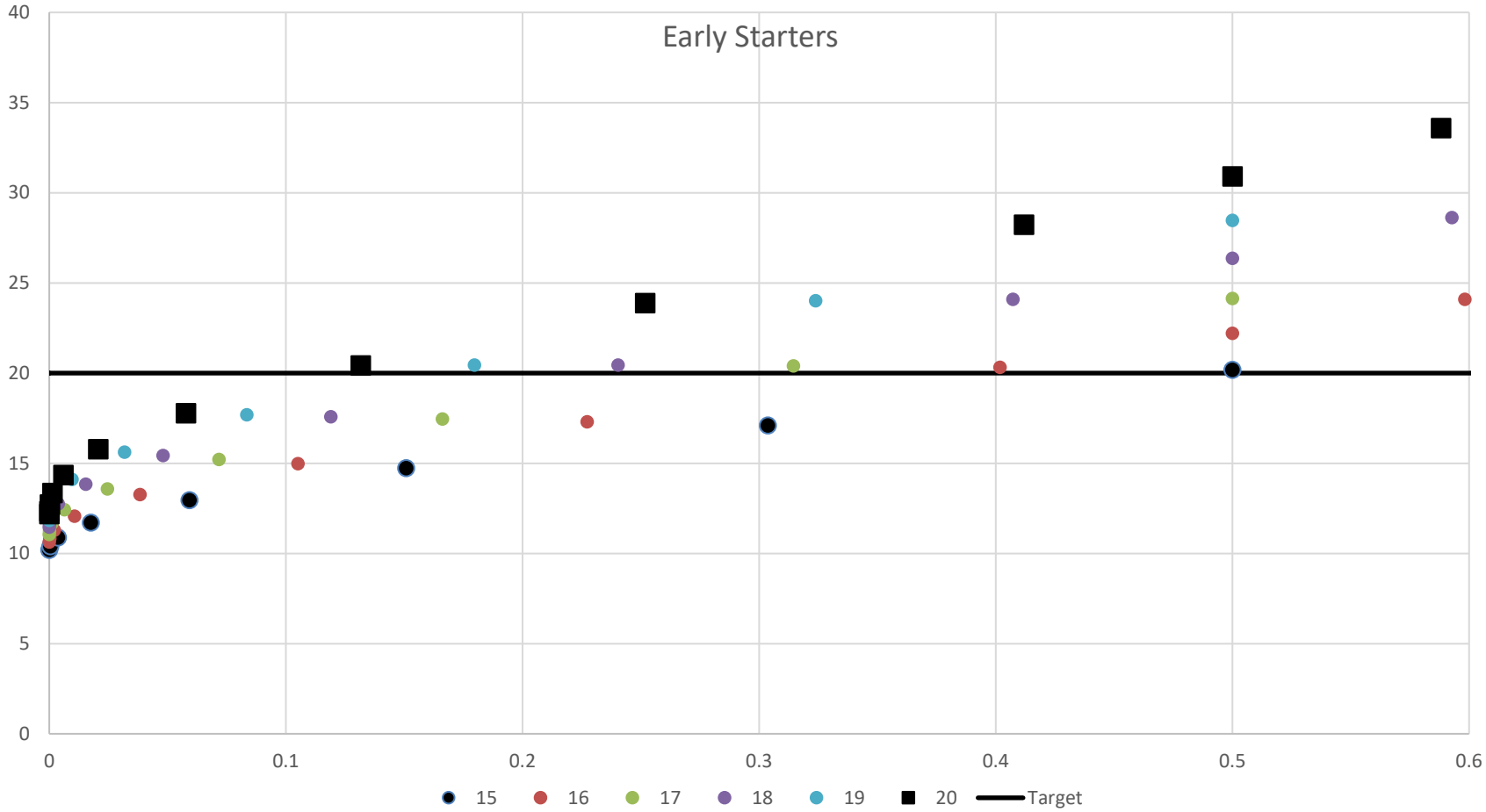
# Solving this problem

- Start at the end
- In the last year invest 1
- get a good draw W= 1.15 (P=0.5)
- Get a bad draw  W=0.95 (P=0.5)
- Back up one year
- 1 invest now two draws
  - two good  $W=1.15^2+1.15=2.475$
  - 1 good 1 bad  $W=(1.25*0.95)+0.5(0.95)+0.5(1.15)=2.195$
  - Two bad  $W=0.95^2+0.95=1.8525$
- Keep iterating backwards


Program 20 years identical investment
Returns g=1.15 b=0.95 prob g=0.5
EV=36
target 20 times initial investment
120
100
80
60
40
20
0
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1
Target
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20




Extremely unlikely to make the target




probability you do not meet the target is now

50% at 15 years 40% at 16 years

32% at 17 years 24% at 18 years

18% at 19 years and 12% at 20 years

# Value at risk

- Need a confidence range: 95%
- Then read off the CDF the value of the program. (easier when distributions are continuous

| | 15 | 16 | 17 | 18 | 19 | 20 |
|---|---|---|---|---|---|---|
| 1% | 11.3 | 11.9 | 13.0 | 13.5 | 16.0 | 15.0 |
| 5% | 12.7 | 13.6 | 14.9 | 15.9 | 16.5 | 17.6 |
| 10% | 13.8 | 13.5 | 16.3 | 16.9 | 18.9 | 19.6 |

Here the investor use the value at risk to decide when to start.
If start 20 years ahead there is 1% chance that the portfolio is worth at less than 15 and 10% chance it's worth at less than 19.6.

# Conditional value at risk

Can use the example to vary the threshold (VaR)

Every Var level corresponds to the realization of certain number of good years

VaR=20.4 => only 7 good years prob 13.2%

CVaR =$\frac{\sum_0^7 (p_i) V_i}{\sum_0^7 (p_i)}$=18.8

VaR=17.8 => only 6 good years prob 5.8%

CVaR =$\frac{\sum_0^6 (p_i) V_i}{\sum_0^6 (p_i)}$=16.8

VaR=13.3 => only 3 good years prob 1.3%

CVaR =$\frac{\sum_0^3 (p_i) V_i}{\sum_0^3 (p_i)}$=13.25

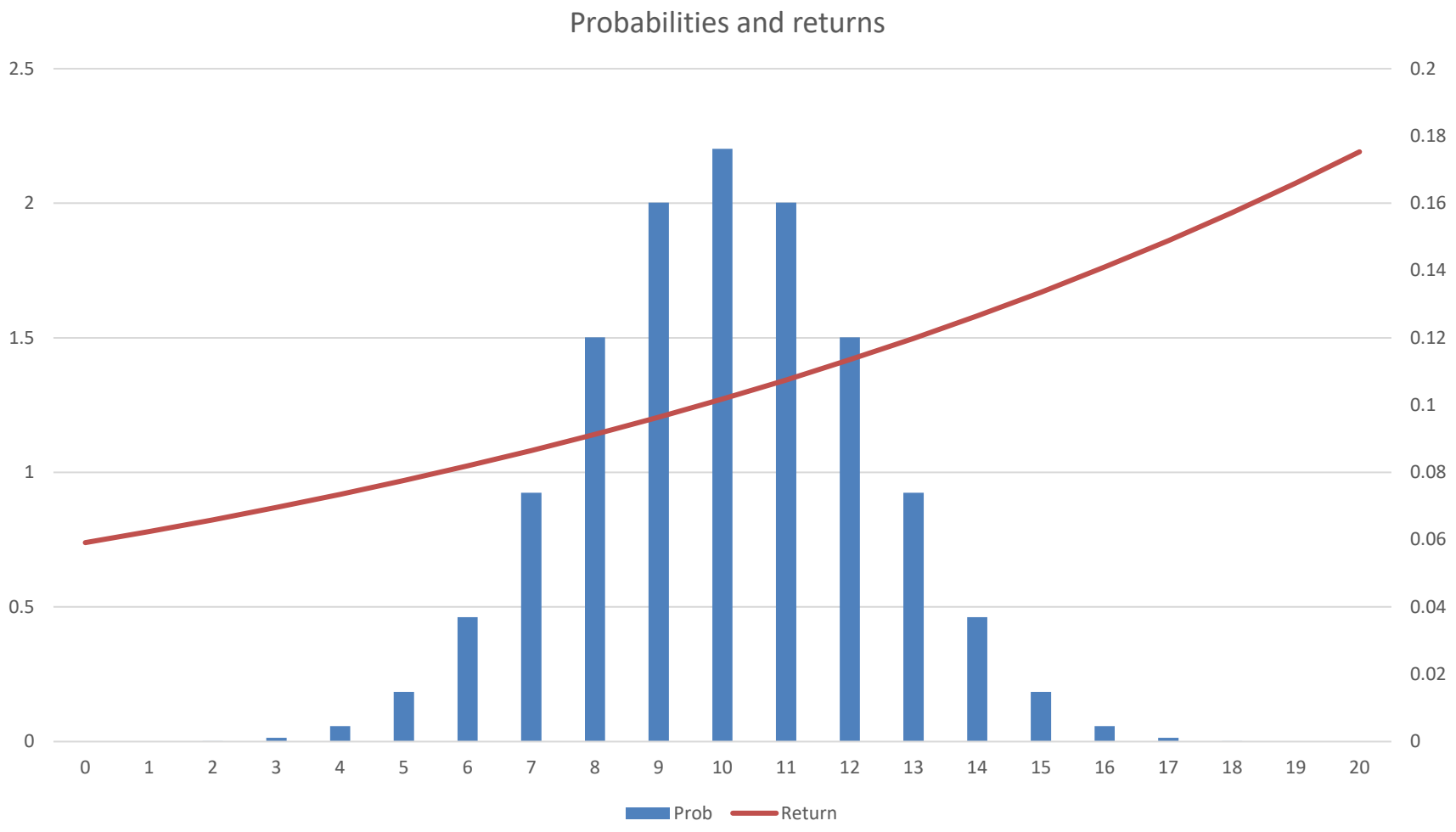
# What to do with this

- So you can use this estimate the likelihood you can retire on time and with what probability you reach your target
- If we look at different programs here just good bad pairs (1.15-95) (125-85) (128-85) can choose what to do.
- Application beyond target decisions

# A fund manager's problem

- Investments fund by individuals who have varying risk tolerances. Negative returns lead to withdrawals
- Suppose the fund is locked in for five year but then half the investors will look at the value of the fund and pull their resources if fund is underwater (worth less than investment).
- The fund reports quarterly g=1.04 b=0.985
- This is a VAR and CVAR problem

# Fund Manager's VAR



# Long Fund pb

- The fund is worth less than investment if there are less than 6 good quarters. This happens with 2.6% probability.
- So fund manager wants to tell clients invest in this fund it has 5.1% expected return and a 2.6% probability of being underwater after 5 years.
- If that occurs its expected value is .951 for an expected loss of 4.8%.
- Manager can take appropriate steps to make sure he can liquidate half the fund in these bad circumstances

# Appropriate steps

- The fund has 5.1% expected return and a 2.6% probability of being underwater after 5 years. With an expected loss of 4.8%.
- One can solve this problem because there is 2.6% probability that the fund will be up at least 69%. With an expected gain of 70.3%
- Sell 7% of the upside to get rid of the downside.
  - For that you need options that are not too expensive
  - And a counterparty

Next time:
Real Estate and Leverage
Problem from
For investors/homeowners
and
For investors/lenders